

Discord Redesign Phase 4

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Abstract— Over the past eight years since its initial release date, Discord has become a prevalent communication social platform in which countless people use on a daily basis. The main functionalities of Discord include instant messaging, voice calls, and video calls for individuals and groups. The main task that I will be discussing involves Discord's group voice channels for group users.

1 QUALITATIVE EVALUATION

For qualitative evaluation, the prototype that will be evaluated will be my paper prototype of design 4. This prototype included the changes of two additions to Discord's current group voice channel interface. The first change to the application will be the inclusion of a small popup to the users whenever he or she is trying to enter or leave a voice channel asking them to confirm the action. The second change is the creation of a small box grid underneath the server name that shows the current channel the user is in as well as other users in the channel. This also removes the user list in a channel from its current location under the server list. The evaluation method for this prototype will be a survey.

1.1 Evaluation Plan

The participants for the qualitative evaluation will be gathered from friends, family, and classmates. An additional requirement for participants of this method is that they must be familiar with Discord or at least have used Discord before. The participants will be recruited through Ed Discussion board or through personal inquiry. Since the evaluation will be in the form of a survey, the process will be asynchronous, which allows the survey takers to do the evaluation on his or her own time and location. There is no restriction in this regard, except for the condition that any feedback should be completely the survey taker's own opinions. Another caveat to utilizing a survey for the qualitative evaluation is that the process will not be recorded since the survey taker has freedom in time and location of the prototype review.

1.2 Layout

- Have you ever joined or left a Discord voice channel unintentionally?
 - 1 - Never
 - 2 - Once or Twice
 - 3 - 3+ times
- Would the popup display in this prototype help or hinder you joining or leaving a Discord voice channel?
 - 1 - Greatly Hinder
 - 2 - Hinder
 - 3 - Neither
 - 4 - Help
 - 5 - Greatly Help
- Would the grid box display help or hinder you joining or leaving a Discord voice channel?
 - 1 - Greatly Hinder
 - 2 - Hinder
 - 3 - Neither
 - 4 - Help
 - 5 - Greatly Help
- Rate your interest in Discord including the popup display?
 - 1 - Very not interested
 - 2 - Not interested
 - 3 - Indifferent
 - 4 - Interested
 - 5 - Very Interested
- Rate your interest in Discord including the grid box display?
 - 1 - Very not interested
 - 2 - Not interested
 - 3 - Indifferent
 - 4 - Interested
 - 5 - Very Interested
- Please describe your thoughts on the popup display
 - Survey taker short answer
- Please describe your thoughts on the grid box display
 - Survey taker short answer
- Please describe any other thoughts on the prototype

- Survey taker short answer
- Please describe any other changes that you think can improve your experience with Discord's group voice channels
 - Survey taker short answer

1.3 Requirements

This evaluation assists greatly in identifying many parts of the data inventory and requirements. The main goal of this design is to improve human error tolerance in the interface for all users. The feedback from the surveys specifically addresses this topic and gives advice on whether or not this issue was alleviated. In addition, the survey shows what the users goals are, what they need, and what are their tasks. The evaluation reflects whether or not the user is able to complete his or her goals through the prototype. In addition, it breaks down the task that the user needs to complete and what he or she needs to complete the task.

2 EMPIRICAL EVALUATION

For the empirical evaluation, the prototype that will be analyzed is the verbal prototype of design 1. This design includes the addition of a "Join" button for users to easily locate how to enter Discord's voice channels. This button will be located right next to the name of the Discord voice channels. In addition, the current buttons that are currently occupying the "Join" button's location will be moved into the right click menu. The empirical evaluation of the verbal prototype will be based on the activity that will be added into the script of the prototype.

2.1 Control and Experimental Conditions

The current verbal prototype incorporates an image of the current Discord interface for the user to reference from. To include an element for empirical evaluation, another image will be added for the user to be able to compare. The image will be of the "Join" button added into the current interface for users to see. The script will also be adjusted to discuss the user's feedback. The test will revolve around how quickly the user is able to identify where to join a Discord voice channel with only one click. Time will be the interval measurement that is used for this evaluation.

2.2 Null and Alternative Hypothesis

- H_{NULL} = The time a new user takes to find the button to click to join Discord's group voice channel is the same in the current interface and the prototype interface
- $H_{\text{ALTERNATIVE}}$ = The time a new user takes to find the button to click to join Discord's group voice channel is the same in the current interface and the prototype interface

2.3 Experimental Method

The experimental method will revolve around the time measurement from the image given during the script conversation. There will be two images, one of the current interface and one of the prototype. The subjects will be subject to both images in a random order to analyze. This follows the within-subjects process of the experimental method. The subjects will all be people who have never used Discord before to avoid users with experience in knowing how to complete the task from memory. In addition, each subject will be randomly given an image to look at first, then the second image to analyze. Both processes will be timed to see how quickly the subject is able to complete the task of finding what to click to join Discord's voice channel. The data generated from this method will be the time that the novice user takes to complete the task, as well as whether or not he or she is able to complete the task. The verbal prototype will then continue with the rest of the unchanged script. Once the data has been gathered, the Student's t-test will be the best analysis process since the dependent variable in this method is an interval measurement. The average times of both subject groups and image orders will be reviewed to determine whether or not the times of the current interface or prototype interface image is within a small enough range to be determined to be the same.

2.4 Lurking Variable

Lurking variables may affect the data that is gathered through this experimental method from the user's side. Although the subjects gathered may be unfamiliar with Discord and are new to the interface, if the user has prior experience with any other application with a similar interface then the user's prior knowledge could affect the time measurement in this method. In addition, since the user will be shown the images back to back, the previous image can also give the user a

hint towards how to complete the same task with the next image. Both of these variables can affect the data that is gathered.

3 PREDICTIVE EVALUATION

For predictive evaluation, the textual prototype will be analyzed based on design 2. Design 2 incorporates in design 1 and adds onto this design by also removing the existing “X” button to leave a voice channel and adds in a “Leave” Button for the user. This button will also be moved to be interchanged with the “Join” Button of design 1, so whenever a user is not in the voice channel the button will display “Join” and when he or she is in the voice channel the button will display “Leave” instead.

3.1 Cognitive Walkthrough

The predictive evaluation will be done through a cognitive walkthrough. The textual prototype design’s requirement is to improve the learnability and user friendliness of the interface. This results in the targeted users being novice users or people who are completely new to Discord. Cognitive walkthrough will be the better task analysis for this prototype compared to the GOMS model.

3.2 Tasks

For this evaluation, there will be two main tasks that are reviewed for analysis, which is joining a Discord voice channel and leaving a Discord voice channel. The user’s goal is to successfully complete a call utilizing the new interface without having to struggle or mess around with the interface to accomplish the tasks. Since the interface is completely digital and does not require any typing, the only operator in this process will be clicking with the computer mouse.

3.3 Evaluation

The predictive evaluation of the textual prototype will be focused on the results of the user’s ability to navigate the interface and complete a voice channel call. Learnability is the goal of improvement for this design so the predictive evaluation will be centered on the interface’s ability to teach a new user and its process in showing the user how to complete the task. In addition, the users’ thought process will also be reviewed to determine whether or not the new interface is able to ease the users’ ability to learn the process of completing the tasks.

4 PREPARING TO EXECUTE

The previous evaluations will be completed in the next assignment, and based on the current evaluation processes on the prototypes, I believe the two best prototypes will be the empirical evaluation and the predictive evaluation. Although the qualitative evaluation can be very useful and gives the most raw data from users compared to the other two evaluations, I believe there can possibly be too much possible biases since the prototype will be using users that have experience with Discord. With the other two methods, the users will either be predictive or a novice user of Discord which can show the best results of whether or not the prototype's interface was able to improve learnability. The predictive evaluation was also chosen for its cognitive walkthrough. The cognitive walkthrough gives me the opportunity to put myself in the user's role to try and perceive how the user will interact with the prototype. Alternatively, the verbal prototype from the empirical evaluation allows me to gain insight on how a novice user would interact with the original interface as well as the prototype. With both designs being closely related, the data gathered from both can help give further feedback in the evaluation process.